Some applications of Monte Carlo Methods in urban climate

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Abstract

In this work we present Monte Carlo Methods as a tool for studying the longwave balance in urban structures. First we make a simple model of an infinite canyon. With this model we test the MCM and obtain a physical explanation of the importance of canyon structure in surface Urban Heat Island formation. Second we elaborate a more complicate model of a street crossing, for analyzing the temperature gradient in that geometry situation. The results obtained are agreement, in a qualitative way, with gauges taken for other authors.

1 Introduction

The Monte Carlo Method (MCM) is a tehcnique of computer simulation of a system with many degrees of freedom. The main idea lies in the use of random numbers to simulate physical systems. These computer experiments are a direct transcription into computing term of a natural stochastic proccess. The use of these kinds of statistical methods allows us to treat many dimensional problems that could not be explored from a deterministic point of view.

There are several characteristics that are necessary to any well

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designed simulation. A clear statement of the system to be simulated is needed. Once this is accomplished, the probability distribution functions that will be involved must be identified and explicitly defined. Finally one must be able to understand how to interpret the information provided by the simulation.

Although Monte Carlo techniques exist since the middle of this century, progress in computers, and of course, faster generation of random numbers have lead to massive use of these techniques.

In this work we propose the use of MCM in the study of urban climate. It is known that the process of urbanization produces radical changes in the surface and atmospheric features of a region. All these changes induce a modification in the climatic balance, creating a new climate, called the urban climate, that must be understood as a local perturbation of the local climate. Of all features of this new climate the Urban Heat Island (UHI) is the most commonly studied. This phenomenon is characterised by a temperature difference between the city and its outskirts.

The structure of this work is as follows: First we shall deal with explicitly a simple geometry in discussing the simulation, and calculate the long-wave radiation loss in a simple urban structure. But such a model includes the more important physical effects that give rise to the Urban Heat Island phenomenon. Second we study the more complicated problem of the gradient surface temperature in a street crossing

2 Radiation balance in urban structures

While the most important phenomenon in urban climatology has been the UHI, the most active debate has been about the identification and specially ranking of its causative factors. However it seems that the effects of street canyon geometry on radiation and the thermal properties on heat storage release, are the primary and almost equal cause on most ocassions. The canyon radiative geometry makes the long-wave radiation loss from within an urban canyon decrease due to the complex exchange between buildings and the screening of the skyline. Therefore we focus our study on this task.

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In this first approach we suppose "ideal conditions" at night because it is when UHI occurs. Also we assume that the walls and ground are black bodies and therefore perfect emitters. Under these conditions, the energy radiated by surface and time is given by the Stephan-Boltzman equation

$$L\uparrow = \sigma T^4 \tag{1}$$

where σ is the Stephan-Boltzman constant. We have not taken into account the possible effects of windows and other feautures of a street, and use the same "materials" for all parts of the canyon. Also we suppose as constant the incoming radiation from the sky.

We consider a finite time step and then we compute the energy loss by volume element, ΔU as the difference between radiation emitted- radiation received. Then the total temperature decrease is

$$\Delta T = \frac{\Delta U}{cV} \tag{2}$$

where c is the heat capacity of material, and V is the volume of each element.

2.1 Problem of infinite canyon

This case can be solved analytically and therefore it will be a good reference for testing the model. We suppose infinite thermal conductivity, that is, temperature is the same for all surfaces of the canyon and therefore all points release the same amount of radiation. Also the incoming radiation from the sky is not taken into account.

The emitted radiation from a differential surface element towards all possible direccions is constant $L_u\uparrow$

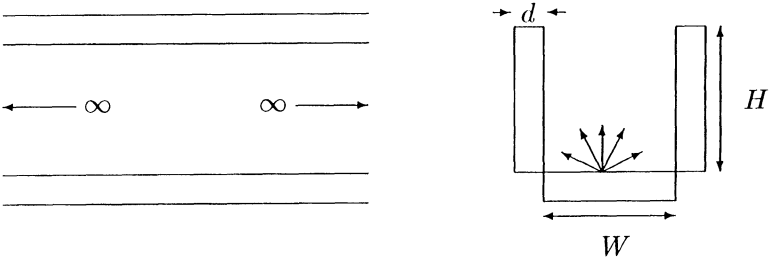


Figure 1: Scheme of the surface canyon model

Let d be the thickness of walls and ground, H the height of the walls, and W the width of the canyon,(see figure 1) then, due to simetry of problem, total emitted radiation is

$$L\uparrow = \int_0^W L dx + \int_0^H L dy \tag{3}$$

And the lost radiation from walls and ground

$$L\uparrow == \int_0^W dx \int_{tan^{-1}(\frac{H}{W-x})}^{\pi - tan^{-1}(\frac{H}{x})} d\phi + 2\int_0^H dy \int_{\frac{\pi}{2}+arctan(\frac{H-y}{W})}^{\pi} d\phi \tag{4}$$

Integrating these expresions we obtain the percentage of radiation loss

$$\frac{(W+H)(\pi - 2tan^{-1}(\frac{H}{W})) + (W-H)ln(W^2+H^2) + Hln(H^2) - Wln(W^2)}{\pi(H+W)} \tag{5}$$

After solving exactly this situation, we are going to reproduce the result by using the MC scheme, in doing so we can test the accuracy of this numerical technique. This is important because it provides us with a measure of the fiability of the method when applied to more realistic situation in which an anlitycal solution is not possible.

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The Monte Carlo simulation consists of generating a fixed number (proportional to T^4) of fotons for each wall and ground. The point of emission and direction is randomly selected. Then the percentage of lost energy is calculated as $\frac{N_{out}}{N_{emitted}}$ where N denotes number of photons.

Looking at figure 2 we can check the good agreement between analytical and Monte Carlo results for several H/W rates. Therefore the MCM is a valid tool to solve the problems we are studying.

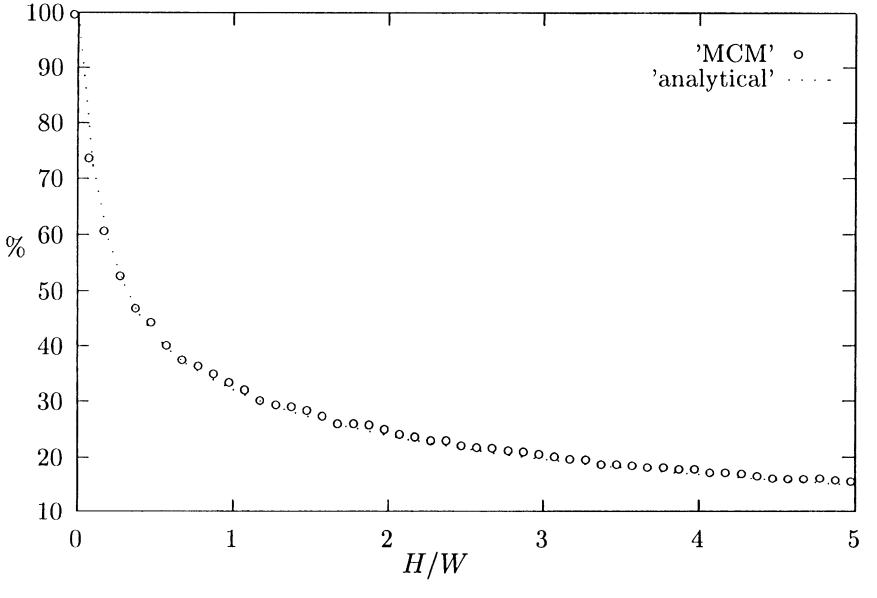


Figure 2: Comparison between MCM and anlytcal models

So, comparing temporal temperature evolution between both an hypothetical rural site ($H = 0$) and an urban canyon, we observe that the canyon is going to cool more slowly than "rural site", which is related to Urban Heat Island formation. The temperature difference is shown in figure 3.

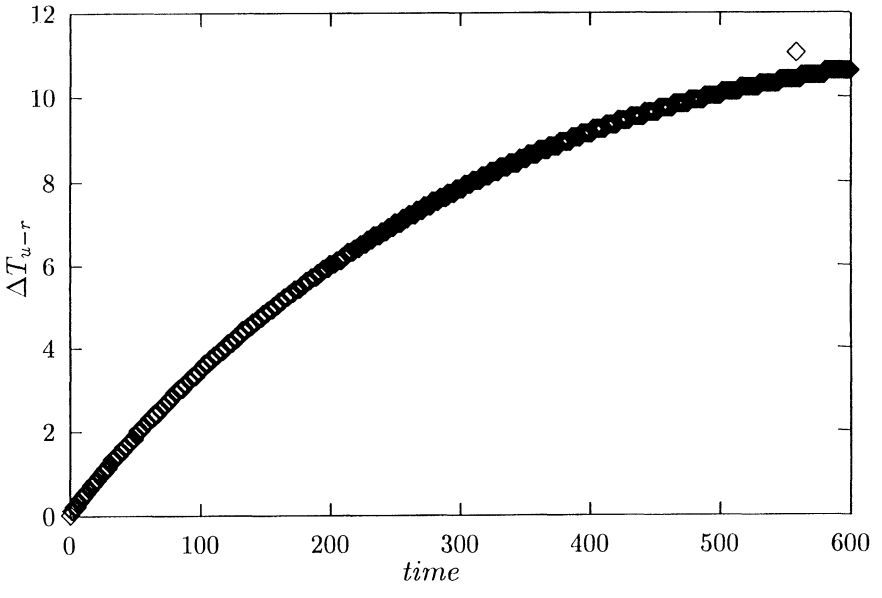


Figure 3: Temporal evolution of temperature difference (K) ΔT_{u-r} between canyon and 'rural site' (time is given in minutes)

2.2 Problem of the crossing

In this section we focuse our attention on a more realistic situation that will give us the main physical effects involved in the temperature surface gradients that ocurr on the street surfaces.

Now the problem to solve is to calculate the temperature gradient in a crossing of streets of infinite length. Due to the simetry of this structure, the problem can be simplified in the way indicated in the figure 4, where L is a finite length of the street, in such a way that in this point we can suppose that the conditions are same as in an infinite canyon. In this way we have gridded walls and ground.

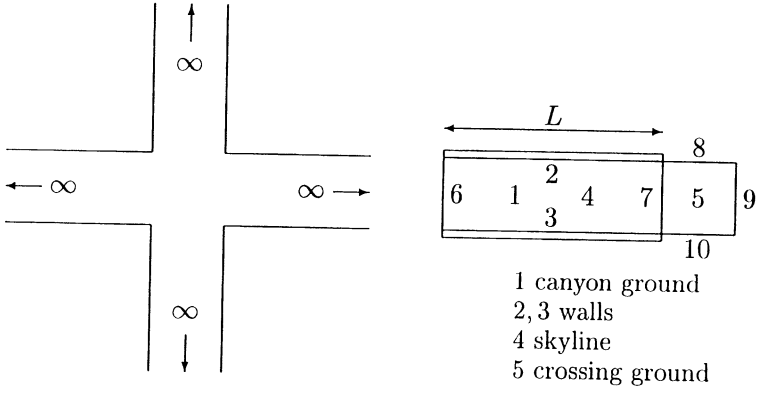


Figure 4: Street crossing and simplification

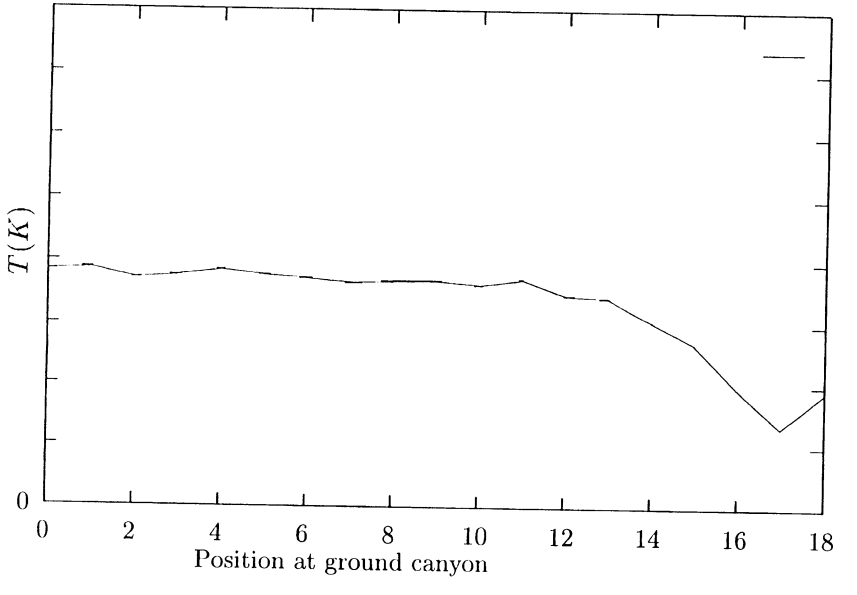


Figure 5: Temperature along the simplified canyon street crossing

So, if the photon impact in:

- 6 is reflected

- 2,3,or 1 the photon is counted in the corresponding grid
- 8,9 or 10, the photon is reemitted from 7 towards the canyon
- 5, if the photon comes from skiline of 5, is counted, and the rest of cases it is counted in four simetrics grids of 5

Afterwards we have calculated the energy loss for each square and therefore the temperature decrease as we have indicated above.

The results obtained can be observed looking at the figure 5, according to the experiments performed by ELiasson, who took surface ground temperature in some streets of Goteborg, obtaining that the air temperature does not have an important variation in crossings while the surface temperature decreases when we approch to the crossings.

3 Comments and conclusions

Although the causes of UHI formation are very complicated and it is very dificult to separate, then this simple model can be a good tool for understanding radiative process in urban canyons, and it can be used to explain the nocturnal surface UHI. This study is only a first approach, and the results are compared with real gauges only in a qualitative way, the complete work is going to develop.

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